MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT202

QP CODE: 25900360 - SENSORS AND ACTUATORS IN INDUSTRY AUTOMATION

Duration: 2 Hours Maximum Marks: 70

Part A

Multiple Choice Questions | Answer all questions | Each question carries 1 marks

Q. No.	Answer	Q. No.	Answer
1.	**(d)**	11.	**(b)**
2.	**(a)**	12.	**(c)**
3.	**(b)**	13.	**(b)**
4.	**(c)**	14.	**(c)**
5.	**(d)**	15.	**(b)**
6.	**(a)**	16.	**(c)**
7.	**(c)**	17.	**(a)**
8.	**(c)**	18.	**(b)**
9.	**(c)**	19.	**(a)**
10.	**(d)**	20.	**(b)**

Part B

Short Answer Questions | Answer any 6 questions | Each question carries 5 marks

1. Explain industrial sensors like temperature or pressure sensor with example.

- **Any two sensors – 2.5 marks each**

Temperature Sensor:

Function: Measures the heat energy or temperature of a system.

Working Principle: Changes in temperature cause changes in voltage, resistance, or current that can be measured electronically.

Examples:

Thermocouple: Generates a voltage based on the temperature difference between two metals. Used in furnaces and boilers.

RTD (Resistance Temperature Detector): Resistance increases with temperature. Common in industrial process control.

Pressure Sensor:

Function: Measures the force exerted by a fluid (liquid or gas) on a surface.

Working Principle: Pressure changes cause deformation in a diaphragm, which is converted into an electrical signal.

Examples:

Piezoelectric Pressure Sensor: Converts pressure into an electric charge. Used in hydraulic systems.

Strain Gauge Pressure Sensor: Detects strain in a diaphragm due to applied pressure. Used in water pump monitoring and pneumatic system.

22. List any five applications of temperature sensors in industry and daily Life.

- **Five applications – 1 mark each**

1. *Industrial Process Control: Monitoring and maintaining temperature in chemical, food, and pharmaceutical industries.*
2. *Boilers and Furnaces: Preventing overheating and ensuring safe operation.*
3. *HVAC Systems: Regulating heating, ventilation, and air conditioning in factories and buildings.*

4. *Refrigeration and Cold Storage: Maintaining required temperatures in freezers and cold rooms.*
5. *Home Appliances: Controlling temperature in ovens, water heaters, and air conditioners.*

23. Compare the advantages and limitations of R–2R ladder DAC

- **Advantages – 2.5 marks**
- **Limitations – 2.5 marks**

Advantages of R–2R Ladder DAC:

1. *Simple Design: Uses only two resistor values (R and 2R), making it easier to design and manufacture.*
2. *Scalable: Can be easily expanded for higher bit resolutions by adding more stages.*
3. *Cost-Effective: Requires fewer precise resistors compared to weighted resistor DACs.*
4. *Stable Performance: Offers better accuracy and linearity compared to a weighted resistor DAC with many bits.*
5. *Easy to Interface: Compatible with digital circuits for converting binary signals to analog voltage.*

Limitations of R–2R Ladder DAC:

1. *Resistor Matching Critical: Accuracy depends on precise R and 2R resistor values.*
2. *Limited Speed: Switching speed may be limited due to parasitic capacitances in high-frequency applications.*
3. *Output Current Limitation: Typically produces voltage output; current output requires additional circuitry.*
4. *Resolution Limit: Practical implementation is limited by resistor tolerance and circuit noise.*
5. *Glitching: Small voltage glitches may occur during transitions between digital codes.*

24. Differentiate between mechanical switches and touch switches.

- **Five differentiation – 1 mark each**

Feature	***Mechanical Switch***	***Touch Switch***
Operation	*Activated by physically pressing or toggling a lever, button, or key.*	*Activated by a simple touch or proximity of a finger.*
Mechanical Parts	*Contains moving parts like springs and contacts.*	*No moving parts; uses capacitive or resistive sensing.*
Durability	*Wears out over time due to mechanical friction.*	*Longer lifespan since no physical wear occurs.*
Response Time	*Slightly slower due to mechanical movement.*	*Fast response; almost instantaneous.*
Applications	*Light switches, push buttons, relays.*	*Modern appliances, smartphones, touch panels, lamps.*
Cost	*Generally cheaper to manufacture.*	*Usually more expensive due to electronic sensing circuitry.*

25. Explain the concept of step angle.

- **Concept of step angle – 2 marks**
- **Explanation – 3 marks**

Concept: A step angle is the angle by which the shaft of a stepper motor moves every time it receives one pulse of electrical input. It defines the smallest rotational movement the motor can make and determines the motor's precision.

Explanation: Step angle = (360°) ÷ (Number of steps per revolution)

Importance: Smaller step angles allow higher positioning accuracy.

Applications: Used in CNC machines, 3D printers, robotics, and precise motion control systems.

26. Describe the working principle of a solenoid.

- **Definition of solenoid – 2 marks**
- **Description – 3 marks**

Definition: A solenoid is an electromechanical device that converts electrical energy into linear mechanical motion.
Explanation: The solenoid works on electromagnetism: when an electric current flows through the coil, it produces a magnetic field. This magnetic field attracts or repels a movable iron plunger inside the coil, causing it to move linearly. The movement can be used to push, pull, or actuate a mechanical device.
Electromagnetic Action: Electric current → magnetic field → motion of plunger.
Linear Motion: Converts electrical signal into straight-line mechanical motion.
Control: Movement occurs only when the solenoid is energized; stops when de-energized.
Applications:
Door locks (electronic locking systems), Valves in water pumps or hydraulic systems, Relays and actuators in automation systems.

27. Explain the concept of smart sensors in the context of IoT. How do they differ from traditional sensors?

- **Concept of smart sensors – 2 marks**
- **Comparison with traditional sensors – 3 marks**

Smart sensors are devices that not only detect physical quantities (like temperature, pressure, or motion) but also process, analyze, and communicate data to other devices over the Internet of Things (IoT) network. They often include embedded microprocessors, memory, and communication modules, allowing them to make decisions locally, reduce data transmission, and enable real-time monitoring.
Difference from Traditional Sensors:

Feature	***Traditional Sensors***	***Smart Sensors***
Function	*Only sense/measure data*	*Sense, process, and communicate data*
Processing	*None (requires external system)*	*Built-in processing capability*
Connectivity	*Usually wired, manual reading*	*Wireless, IoT-enabled*
Decision-making	*None*	*Can trigger actions or alerts locally*
Data Handling	*Raw data output*	*Filtered, analysed, or formatted data*

28. What is an LDR sensor, and how does it work for measuring light intensity?

- **Definition – 2 marks**
- **Working – 3 marks**

LDR Sensor (Light Dependent Resistor):
An LDR sensor is a type of sensor whose resistance changes according to the intensity of light falling on it. It is widely used to measure light levels in devices like street lights, brightness meters, and automatic lighting systems.
Working:
The LDR is made of a semiconductor material that has high resistance in darkness and low resistance in bright light.
When light falls on the LDR, photons excite electrons in the material, reducing its resistance.
This change in resistance can be measured by a circuit (like a voltage divider) and converted into a corresponding light intensity value.

Part C

Essay Questions | Answer any 2 questions | Each question carries 10 marks

29. Explain the application of IR sensor in proximity detection and obstacle avoidance with neat diagram.

- **Basics of IR sensor – 2 marks**
- **Proximity detection – 3 marks**
- **Obstacle avoidance – 2 marks**
- **Diagram – 3 marks**

An IR sensor detects objects or obstacles by emitting infrared light and measuring the reflected light from nearby objects. It is widely used in proximity detection and obstacle avoidance in robotics, automation, and security systems.
Working Principle for Proximity Detection and Obstacle Avoidance:
Emitter and Receiver:
The IR sensor has an IR LED (emitter) that sends out infrared light. It also has a photodiode or phototransistor (receiver) to detect the reflected IR light.
Detection Mechanism: When an object comes within the sensor's range, the emitted IR light reflects off the object.
The receiver detects this reflected light and produces an electrical signal. This signal can trigger actions like stopping a robot, changing direction, or sounding an alert.
Obstacle Avoidance: In robots, multiple IR sensors can be placed at the front and sides.
When the sensors detect an obstacle, the microcontroller decides a path to avoid collision, enabling autonomous navigation.
Applications: Automatic street lights, Robotic obstacle avoidance, Proximity alarms, Touchless switches.

30. Describe the different types of microphones with neat sketches.

- ***Principle of microphone – 2 marks***
- ***Any four types of microphone with sketch – 8 marks***

Principle: Microphone is a transducer which converts sound (acoustic energy) → electrical signal.
Main types (based on working principle):
1. Dynamic (Moving Coil) Microphone
Principle: Electromagnetic induction
Diagram
Working: Sound pressure vibrates diaphragm → coil moves in magnetic field → voltage induced.
Features: Rugged, no extra supply, used in stage/PA systems.
2. Condenser (Capacitor) Microphone
Principle: Change in capacitance
Working: Diaphragm + back plate form capacitor. Sound changes spacing → capacitance changes → converts to voltage (need DC bias supply called phantom power).
Features: very sensitive, wide frequency response, used in studio recording
3. Electret microphone
Principle: Condenser but with permanently charged electret material
***Working:** Electret keeps static charge permanently → no need high external bias. Contains small FET inside to buffer output.*
Features: Cheap, used in mobiles, hearing aids, embedded systems
4. Ribbon Microphone
Principle: Induction using thin metal ribbon
Sketch
Sound → Vibrates thin metal ribbon suspended inside magnet gap → EMF
Features: warm, smooth response, delicate, used in broadcast studios.
5. Piezo-electric Microphone
Principle: Piezo effect
Sound → pressure on Piezo crystal → generates voltage
Features: simple, low cost, used in toys, contact pickups of musical instruments.

31. Explain the fundamental working principle of a servo motor, emphasizing its closed-loop control and feedback mechanisms.

- ***Basic operation/Purpose of servo motor – 2 marks***
- ***Block level working principle – 5 marks***
- ***Feedback mechanism – 3 marks***

A servo motor is a special type of motor which operates on closed loop control. It continuously compares its actual position with the commanded position and adjusts itself automatically until the error becomes zero.
Block level working principle
A servo system consists of:
1. Reference/Command Signal (Input)
2. Error Detector / Controller (PID/Drive)
3. Power amplifier / driver
4. Servo Motor (DC, BLDC or AC type)
5. Feedback device – Potentiometer / Encoder / Resolver
Working:
1. The required position/angle/speed is given to the servo as reference input.
2. The actual position of shaft is measured through a feedback sensor (mostly encoder or potentiometer).
3. The controller compares reference input and feedback signal → Error = Reference – Actual
4. The error signal is processed (PID) and given to the power amplifier.
5. The motor rotates in such a direction to reduce the error.
6. As the shaft reaches the desired position, the feedback becomes equal to reference → error becomes zero.
7. At zero error → motor stops instantly and holds the position.
Closed Loop Control:
Servo is NOT open loop like normal DC motors.
It always works in closed loop:
→ It continuously checks "Where am I now?" using feedback
→ It continuously corrects itself to reach the target value
Therefore servo can achieve high accuracy, fast response, precise positioning and holding torque.
Feedback mechanism:
Potentiometer gives an analog voltage proportional to shaft angle.
Encoder gives pulses per revolution (digital feedback).
Resolver gives sine/cosine analog signals representing shaft position. This feedback is the heart of servo principle.

32. Discuss how smart sensors can be implemented using ESP32 and ThingSpeak in an IoT system.

- ***Implementation – 4 marks***
- ***Role of ESP32 – 3 marks***
- ***Role of ThinkSpeak – 3 marks***

In IoT based monitoring systems, smart sensors are created by combining normal sensors with a Wi-Fi enabled controller and cloud analytics. ESP32 and ThingSpeak are common low-cost solutions to implement such smart sensor systems.
System Structure
1. Sensor – temperature, humidity, gas, soil moisture etc.
2. ESP32 – reads sensor signal, converts it to usable value, applies logic and connects to Wi-Fi.
3. Internet / Wi-Fi Router
4. ThingSpeak Cloud – stores sensor data, displays graphs and performs analytics.
Working:
Sensors are interfaced to analog/digital pins of ESP32.
ESP32 reads the sensor output periodically.
ESP32 performs local processing like filtering, calibration, averaging or threshold detection → this makes the normal sensor into a smart sensor.
ESP32 connects to Wi-Fi and sends the processed data using HTTP/MQTT API to ThingSpeak server.
ThingSpeak stores readings in channels and plots live graphs.
ThingSpeak can also use MATLAB analytics to process data, predict trends, and generate alarms.